

Inequalities

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The Symbols $>$, $<$, $\geq$, $\leq$ are Inequalities.

e.g. $8 > 7$, $9 > 2$, $x > -\frac{2}{5}$

Key points

1. Adding or Subtracting the same quantity from both sides of an inequality leaves Symbol unchanged.

e.g. If $8 > 5$, then $\Rightarrow 8 + k > 5 + k$

2. Multiplying or dividing both sides by a positive number leaves the inequality unchanged.

3. Multiplying or dividing both sides by a negative number reverses the inequality

e.g. If $8 > -5 \Rightarrow$ Multiplying both sides with -1 we have.

$$(-1)(8) < (-1)(-5) \Rightarrow -8 < 5 \text{ or } 5 > -8$$

A Letter used to represent any element of a given Set is called a variable.

Let us now look at inequalities that involve variables.

Example 1:

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Solve the inequality

$$4x + 3 > 2x - 5$$

Solution

$$4x + 3 > 2x - 5$$

$$-3 + 4x + 3 > 2x - 5 - 3 \quad \text{[Subtracting 3 on both sides]}$$

$$4x > 2x - 8$$

$$4x - 2x > -8$$

$$2x > -8 \Rightarrow x > -4$$

$\Rightarrow$ The solution consists of all real numbers greater than -4 i.e. $(-4, \infty)$

Example 2:

Solve the inequality.

$$-5 < \frac{4 - 3x}{2} < 1$$

Solution

$$-5 < \frac{4 - 3x}{2} < 1 \Rightarrow$$

$$-10 < 4 - 3x < 2 \quad \text{[Multiplying both sides by 2]}$$

$$\Rightarrow -14 < -3x < 2 - 4 \quad \text{[Subtracting 4 on both sides]}$$

$$-14 < -3x < -2$$

$$\frac{14}{3} > x > \frac{2}{3} \quad \text{[Dividing each side by -3 reverses the inequality]}$$

Example 3.

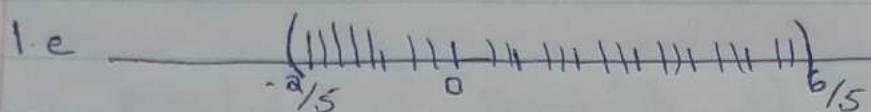
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Solve the inequality $|5x-2| < 4$.

Solution

Recall that $|x| < a \Rightarrow -a < x < a$.

$$\begin{aligned}\Rightarrow |5x-2| < 4 &\Rightarrow -4 < 5x-2 < 4 \\ -4+2 &< 5x < 4+2 \\ -2 &< 5x < 6 \\ \frac{-2}{5} &< x < \frac{6}{5}\end{aligned}$$



Example 4:

Solve the inequality $|1-2x| < 5$.

Solution

$$\begin{aligned}|1-2x| < 5 &\Rightarrow -5 < 1-2x < 5 \\ -5-1 &< -2x < 5-1 \\ -6 &< -2x < 4 \\ 3 &> x > -2 \\ -2 &< x < 3.\end{aligned}$$

Example 5.

Solve the inequality $|7x-3| > 1$.

Solution

Recall that $|x| > a \Rightarrow x > a$ or $x < -a$

$$\Rightarrow 7x-3 > 1 \quad \text{or} \quad 7x-3 < -1$$

$$7x > 1+3 \quad \text{or} \quad 7x < -1+3$$

$$7x > 4 \quad \quad \quad 7x < 2$$

$$x > \frac{4}{7} \quad \quad \quad x < \frac{2}{7}$$

$$\Rightarrow x > \frac{4}{7} \text{ or } x < \frac{2}{7} \quad \text{i.e. } (-\infty, \frac{2}{7}) \cup (\frac{4}{7}, \infty)$$

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Exercise:

Solve the following inequalities.

$$1. |2x+1| > 3 \quad \text{Ans: } x > 1 \text{ or } x < -2.$$

$$2. |1-5x| > 2 \quad \text{Ans } x < -\frac{1}{5}, x > \frac{3}{5}.$$

$$3. |5x| < 1 \quad \text{Ans } -\frac{1}{5} < x < \frac{1}{5}.$$

$$4. 2+5x \geq 1 \quad \text{Ans } x \geq -\frac{1}{5}.$$

$$5. \left| \frac{7-3x}{2} \right| \leq 1 \quad \text{Ans. } \left[\frac{5}{3}, 3 \right]$$

$$6. 3x^2 + 5x - 2 < 0 \quad \text{Ans. } (-2, \frac{1}{3})$$

Complex Numbers:

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NB: The set of real numbers is closed under Multiplication, Addition, Subtraction and division.

Question: What are the real roots of the Quadratic equation $x^2 + 1 = 0$?

You realize that within the real number system, the equation $x^2 + 1 = 0$ has no roots since there is no real number x such that $x^2 = -1$ or $x = \sqrt{-1}$.

To solve this issue, complex numbers were introduced.

Definition: Any number of the form

$z = x + iy$ where $x, y \in \mathbb{R}$ is called a complex number, where

i is an imaginary unit such that $i^2 = -1$.

x is the real part of z denoted by $\text{Re}(z)$.

y is the imaginary part denoted by $\text{Im}(z)$.

$\Rightarrow$ Complex number has two parts, the real part and imaginary part.

e.g. $z = 5 - i2$, then $\text{Re}(z) = 5$
 $\text{Im}(z) = -2$.

Remark:

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(i) The number $0+ib$, $b \in \mathbb{R}$ is known as a pure imaginary number.

(ii) Any real number x is also a complex number and is written as $x+io$.

The Set of complex numbers is denoted by

$$\mathbb{C} = \{Z: Z = x+iy : x, y \in \mathbb{R}\}$$

$$\Rightarrow \mathbb{R} \subset \mathbb{C}$$

Basic Arithmetic operations on Complex Numbers

Let $z_1 = a+ib$ and $z_2 = c+id : a, b, c, d \in \mathbb{R}$.

1. Addition:

$$\begin{aligned} z_1 + z_2 &= (a+ib) + (c+id) \\ &= a+ib+c+id \\ &= (a+c) + ib+id \\ &= (a+c) + i(b+d) \end{aligned}$$

Which is also a complex number.

2. Subtraction:

$$\begin{aligned} z_1 - z_2 &= (a+ib) - (c+id) \\ &= a+ib-c-id \\ &= (a-c) + i(b-d) \end{aligned}$$

Which is also a complex number.

3. Multiplication:

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$$z_1 \times z_2 = (a+ib)(c+id)$$

$$ac + iad + ibc + i^2 bd$$

$$\text{but } i^2 = -1 \Rightarrow i^2 bd = -bd.$$

$$\Rightarrow z_1 \times z_2 = (ac - bd) + i(ad + bc)$$

Which is also a complex number.

Example:

$$\text{Let } z_1 = -4 + 6i \text{ and } z_2 = 3 - 5i$$

Find

$$(i) z_1 + z_2$$

$$= (-4 + 6i) + (3 - 5i)$$

$$= (-4 + 3) + (6 - 5)i$$

$$= -1 + i$$

$$(ii) z_1 - z_2$$

$$= (-4 + 6i) - (3 - 5i)$$

$$= -4 - 3 + 6i + 5i$$

$$= -7 + 11i$$

$$(iii) z_1 \times z_2$$

$$= (-4 + 6i)(3 - 5i)$$

$$= -12 + 20i + 18i - 30i^2 \text{ (but } i^2 = -1)$$

$$= -12 + 38i + 30$$

$$= 18 + 38i$$

Before we go on to the quotient of two complex numbers, we define the conjugate of a complex number as it will help us carry out the operation of division.

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Let $z = a + bi$, we define the conjugate of z denoted by $\bar{z}$ to be the number $a - bi$ i.e. $\bar{z} = a - ib$.

eg if $z = -4 + 6i$, then $\bar{z} = -4 - 6i$

$z = 2 - 3i$, then $\bar{z} = 2 - (-3i) = 2 + 3i$

The pair z and $\bar{z}$ have the following properties

(a) $z + \bar{z} = (a + ib) + (a - ib) = 2a = 2 \operatorname{Re}(z)$

(b) $z - \bar{z} = (a + ib) - (a - ib) = 2bi = 2i \operatorname{Im}(z)$

(iii) $z \times \bar{z} = (a + ib)(a - ib) = a^2 + b^2$, a real number.

Now, let us go back to the operation of division

4. Division:

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$, then

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{(a + ib)}{(c + id)} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} \\ &= \frac{(ac + bd)}{c^2 + d^2} + \frac{(bc - ad)}{c^2 + d^2}i \end{aligned}$$

$\Rightarrow$ The quotient is also a complex number.

Remark: $\mathbb{C}$ is closed under the operations of $+$, $-$, $\times$, and $\div$.

Laws of Arithmetic operations on $\mathbb{C}$

(i) Commutative:

For any two complex numbers z_1 and z_2

$$z_1 + z_2 = z_2 + z_1$$

2. Associative

For any three complex numbers z_1, z_2 & z_3
 $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$.

3. Distributive

For any three complex numbers z_1, z_2 & z_3 .
 $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$.

Square Roots of Complex numbers:

For Example, find the square roots of

(i) $2i$ (ii) $5 - 12i$

Solution

Let $2i = z^2$ Since it is the square of some complex number z .

But we know that $z = x + iy; x, y \in \mathbb{R}$.

$$\Rightarrow z^2 = (x + iy)^2$$

$$\Rightarrow (x + iy)^2 = 2i$$

$$x^2 + 2xyi - y^2 = 2i$$

$$(x^2 - y^2) + 2xyi = 2i$$

Equating the real and imaginary parts, we get

$$x^2 - y^2 = 0 \quad \text{and} \quad 2xyi = 2i$$

from $x^2 - y^2 = 0 \Rightarrow x^2 = y^2 \Rightarrow x = y$. ①

from $2xyi = 2i \Rightarrow 2xy = 2 \Rightarrow xy = 1 \Rightarrow x = \frac{1}{y}$. ②

Since from ① we have $x = y$

$$\Rightarrow \frac{1}{y} = y \Rightarrow y^2 = 1$$

$$y = \pm 1$$

If $y = 1$, then $x = 1$, If $y = -1$, then $x = -1$.

The square root of $2i$ is $1+i$ and $-1-i$.

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(ii) Let $z^2 = 5 - 12i$, then if $z = (x + iy)$, $z^2 = (x + iy)^2$

$$\Rightarrow (x + iy)^2 = 5 - 12i$$

$$x^2 + 2xyi - y^2 = 5 - 12i$$

$$(x^2 - y^2) + 2xyi = 5 - 12i$$

$$\Rightarrow x^2 - y^2 = 5 \quad \text{and} \quad 2xyi = -12i$$

$$\text{From } 2xyi = -12i \Rightarrow y = -6/x$$

Substituting that in $x^2 - y^2 = 5$, we obtain

$$x^2 - \left(-\frac{6}{x}\right)^2 = 5 \Rightarrow x^4 - 5x^2 - 36 = 0$$

$$(x^2 + 4)(x^2 - 9) = 0$$

$$\Rightarrow x^2 = -4 \text{ or } x^2 = 9, \text{ since } x \text{ is a real number}$$

$$x \neq \sqrt{-4}, \text{ thus } x^2 = 9 \Rightarrow x = 3 \text{ or } x = -3.$$

$$\text{When } x = 3, y = -2$$

$$x = -3, \text{ then } y = 2.$$

$\Rightarrow$ The square roots of $5 - 12i$ are $3 - 2i$ and $-3 + 2i$.

Exercise:

Find the square roots of the complex number

(i) $-8i$

(ii) $35 + 12i$

(iii) $-15 - 8i$

(iv) $3 + 4i$

Absolute Value of a Complex Number (23)

The absolute value or modulus of a complex number $z = x + iy$ is defined as the non-negative number $\sqrt{x^2 + y^2}$ and is denoted by $|z|$ i.e

$$|z| = \sqrt{x^2 + y^2}$$

Examples:

Find the modulus of the following complex numbers

(i) $-3 + i4$

(ii) $1 + 4i$

(iii) $-1 - i$

(iv) -8

Solution

(i) $|-3 + 4i| = \sqrt{(-3)^2 + 4^2} = 5$

(ii) $|1 + 4i| = \sqrt{1^2 + 4^2} = \sqrt{17}$

(iii) $|-1 - i| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$

(iv) $|-8| = |-8 + i0| = \sqrt{(-8)^2 + 0^2} = 8$

Exercise

1. Find the modulus of the following complex numbers

(i) $4i$

(ii) $-\sqrt{3} + i$

(iii) $-6 - 6i$

(iv) $\frac{(-2 + i)(8 + 4i)}{5 - 3i}$

2. Given that $z_1 = (1-i)(1+2i)$

$$z_2 = \frac{2+6i}{(3-i)}$$

$$z_3 = \frac{-4i}{1-i}$$

(i) Express each of these complex numbers in the form $x+iy$, where $x, y \in \mathbb{R}$.

(ii) Show that $|z_1 - z_3| = |z_1 - z_2|$

(iii) Find $\left| \frac{z_1 + 2z_2 - z_3}{i4 - 3z_1} \right|$.

3. If $z = 1+i$, find the real and imaginary parts of

(i) $z + \frac{1}{z}$

(ii) $\frac{z+1}{z-1}$

4. If $a = 3-i$ and $b = 1+2i$, find the modulus of

(i) $2a+b$

(ii) $\frac{a}{2b}$